

VIDIT AGARWAL

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About

10+ years experienced, meticulous & detail-oriented machine learning engineer & deep learning expert possessing a proven track record of deploying artificial intelligence components for processing voluminous datasets & extracting insights to facilitate sound decision-making by management. Adept at optimizing basic and convolutional neural networks & employing deep learning methodologies to ensure effective strategy formulation & implementation. Executing full life-cycle development projects; ramping up projects within time, budget & quality parameters, as per project management & best practice guidelines, targeting assignments in **Data Science, Machine Learning, Data Engineering and Artificial Intelligence** with an organization of high repute. Highly skilled in various ML/DL techniques, & predictive analysis to successfully execute projects and deliver compelling business value. Proficient in querying and manipulating data using Python and SQL. Possesses an excellent understanding of complex machine learning and statistical modeling algorithms/techniques for extracting valuable insights and enhancing customer satisfaction.

SUMMARY

- Achievement-driven professional with an experience of nearly 10+ years
- Experience in architecting Artificial Intelligence applications with Machine Learning, Deep Learning with Python. Deep Learning Techniques includes ANN, CNN, RNN, Advanced CNN and Advanced RNN with LSTM.
- Skilled in libraries such as Pytorch, PySpark, Keras, Tensorflow, Sklearn, Numpy, Pandas, Matplotlib, Seaborn, Tableau for Data Visualization.
- Expertise in manipulating and analyzing complex, high-volume, high-dimensionality data from varying data sources and big data sources.
- Hands on expertise with NLP specifically with latest tech Chat GPT, LLM's, Haystack, Hugging Face, Cognitive services in clouds.
- Expertise on most of the useful Cloud Tools on AWS, AZURE as well as GCP.
- Expertise building Live Dashboards, proficient in using Data Studio, Looker, Qlik Sense, Power BI, Tableau.
- Hands on experience in Computer Vision Technologies with openCV, pytorch, tensorflow, Scikit-learn.
- Corporate Training Experience for more than 3.5+ years.

TECHNICAL SKILLS

Packages: SciKit-Learn, NumPy, SciPy, Plot.ly, Pandas, NLTK, BERT, Matplotlib, StatsModels, Lambda, AWS glue, Redshift, EC2, GraphQL, Sharepoint, Python, ARIMA, SARIMAX, ARMA, Microservices, Jupiter Notebook, Time-Series, TensorFlow, Pytorch, Kotlin

Big Data Stack: Hadoop, Apache Pig, Python, PostgreSQL, AWS, Hive, MongoDB, MapReduce, Spark, Linux, R, SQL, SAS, Power BI, Tableau

Statistics/ML: Linear/Logistic Regression, SVM, Ensemble Trees, Random Forests, Clustering, Markov Model, Gradient Boosted trees, Sagemaker, SAP HANA ML, LLM's, Haystack, OpenAI, BERT, CHAT GPT-3, 4, Vertex.AI, Sagemaker, Azure ML, MLflow, KubeFlow, Databricks

Visualisation/Analytics Tools: PowerBI, Tableau, QuickSight, Data Studio, ER Studio

Clouds: AWS, GCP, AZURE, Talend, SAP HANA

PROFESSIONAL EXPERIENCE

Lead Full Stack Data Scientist and Machine Learning Consultant

Dec '18 - Present

Ecosystem Consulting | Remote, IN

Ecosystem Consulting is a start-up Consulting which specializes in delivering solutions to clients from different domains

Technology Stack: Python, SAP HANA, Looker, Hadoop, Power BI, Pandas, NumPy, SAP HANA ML, SciKit-Learn, Lambda, Tableau, AWS Glue, RedShift, Deep learning Reinforcement and NLP, GCP, pyspark, Scala, Sagemaker, Titan, AWS cloud, GraphQL, Rest, Jupiter, Langchain, Typescript, Notebook, BERT, OpenAI, Vertex.AI, Haystack, LLM's, Databricks, Vector Database

Project1: Voice-based Query Bot

Role: Gen AI Consultant

Duration: Aug/2023 – May/2024

Description: Developed a voice-based query bot aimed at enhancing user engagement and experience during the G20 Summit 2023. The solution utilized advanced AI technologies to provide accurate and reliable information through voice interactions.

Key Responsibilities:

- **Project Planning:** Developed comprehensive project plans outlining timelines, resources, and deliverables.
- **Technology Integration:** Leveraged Azure services including Cognitive Search, Speech-to-Text, Text-to-Speech, and GPT-3 for model training and inferencing.
- **System Architecture:** Designed and implemented solution architecture incorporating Azure Machine Learning, Azure Blob Service, and Azure Redis.
- **Development and Testing:** Oversaw the end-to-end development and rigorous testing of the integrated system to ensure performance and reliability.
- **Collaboration:** Coordinated with cross-functional teams, including data scientists, ML engineers, and frontend developers to meet project milestones.

Achievements:

- Successfully deployed a multilingual bot capable of handling voice inputs in various languages.
- Enhanced user engagement by providing accurate, real-time responses to trade-related queries.
- Delivered the project within the stipulated timeline and budget, ensuring high client satisfaction.
- Achieved seamless integration with Azure services, providing a robust and scalable solution.
- Leveraged **LangChain** to build robust and scalable chains of transformers for various NLP tasks.
- Integrated vector databases for efficient storage and retrieval of high-dimensional data, enhancing the performance of AI models in similarity search and recommendation systems.
- Employed vector databases like FAISS and Pinecone for fast and scalable similarity searches.

Technologies Used:

- **Cloud Platform:** Microsoft Azure
- **Services:** Azure Cognitive Services, Azure Speech Services, Azure Cognitive Search, Azure Machine Learning
- **AI/ML:** GPT-3, Neural Voice Pipeline, LLM's RAG's, Prompt Engineering
- **Programming Languages:** Python, Sql, pyspark, scala, Javascript
- **Tools:** Power BI, Azure DevOps, Microsoft Teams

Libraries and APIs: OpenAI API, LUIS (Language Understanding Intelligent Service) - Microsoft Azure, Typescript, LangChain.js, Next.js

Project2: Development of Online Proposal Evaluation System for DEWA

Role: Gen AI Consultant

Duration: Sep/2022 - Aug/2023

Description: Led the development of an online system to streamline the proposal evaluation process for the Dubai Electricity and Water Authority (DEWA). The system aimed to improve efficiency, standardization, and accuracy in handling technical and commercial proposals.

Key Responsibilities:

- **Project Management:** Planned, executed, and monitored project timelines, ensuring the project was delivered on time and within budget.
- **System Design:** Designed the architecture of the online proposal evaluation system using Microsoft Azure services.

- **Integration:** Integrated various Azure services such as Azure Cognitive Services, Azure Machine Learning, Azure Blob Storage, and Azure Cosmos DB.
- **Development:** Led a team of developers and data scientists to build and deploy the system, ensuring high performance and scalability.
- **NLP and LLM Implementation:** Utilized Natural Language Processing (NLP) techniques and Large Language Models (LLMs) from OpenAI, such as GPT-3, for processing and understanding proposal documents, as well as generating accurate and relevant responses.
- **API Integration:** Implemented and integrated OpenAI APIs for advanced language understanding and generation tasks.
- **Testing and Validation:** Conducted rigorous testing and validation to ensure the system met all requirements and specifications.
- **Stakeholder Communication:** Facilitated communication between DEWA and vendors through post-tender clarifications and generated comprehensive evaluation reports.

Achievements:

- Successfully developed a unified platform that reduced the proposal evaluation time by 45%.
- Enhanced the accuracy and consistency of proposal evaluations through automated validation and rating systems.
- Improved collaboration and communication between DEWA and its vendors.
- Delivered a robust, scalable solution leveraging advanced Azure services and OpenAI technologies.
- Leveraged **LLMs** for context-aware document generation and retrieval.
- Utilized **LangChain's modular architecture** to streamline the development process and improve maintainability.
- Utilized prompt engineering techniques to refine model responses and ensure high-quality output.
- Fine-tuned pre-trained **RAG** and **LLM models** on domain-specific datasets to improve relevance and accuracy.
- Conducted extensive A/B testing to validate the performance improvements of the integrated system.
- Implemented graph databases to model and analyze complex relationships between entities, enabling sophisticated querying and insights.
- Utilized graph databases like Neo4j to support advanced analytics and knowledge graph applications.

Technologies Used:

- **Cloud Platform:** Microsoft Azure
- **Services:** Azure Cognitive Services, Azure Machine Learning, Azure Blob Storage, Azure Cosmos DB, Azure App Service
- **AI/ML:** GPT-3 for model training and inferencing, leveraging OpenAI's NLP capabilities, LLM's RAG's, Prompt Engineering
- **Programming Languages:** Python, Sql, Pyspark, Scala
- **Tools:** Azure DevOps, Power BI
- **Libraries and APIs:** OpenAI API, RAG's, Typescript, LangChain.js, Next.js, Rasa NLU

PROJECT3: STOCK PRECITION CHATBOT BASED ON GEN-AI

Overview: Developed an intelligent chatbot for providing real-time stock market information and personalized investment advice using generative AI. The chatbot integrates with financial data sources and uses advanced NLP techniques to understand and respond to user queries.

Key Responsibilities:

- **Project Planning and Requirements Gathering:** Collaborated with financial analysts and stakeholders to define project requirements and use cases.
Designed a project roadmap and timeline, ensuring alignment with business goals.
- **Data Collection and Integration:** Integrated APIs from multiple financial data providers to gather real-time stock market data.
Ensured data accuracy and consistency by implementing data validation and error-handling mechanisms.
- **Model Development:** Developed a natural language processing pipeline using GPT-3 to generate coherent and contextually relevant responses.
Trained the model on a diverse dataset of financial news articles, stock market reports, and historical trading data to enhance its understanding of financial terminology and concepts.

Implemented reinforcement learning techniques to improve the model's performance based on user feedback.

- **Chatbot Development and Deployment:** Designed and developed the chatbot interface using React and Node.js. Deployed the chatbot on AWS, ensuring scalability and reliability through load balancing and auto-scaling configurations. Implemented a continuous integration and deployment (CI/CD) pipeline to streamline updates and maintenance.
- **User Experience and Feedback:** Conducted user testing sessions to gather feedback on chatbot performance and user experience. Improved chatbot accuracy and user satisfaction by iteratively refining the model and interface based on user feedback.

Achievements:

- Architected and implemented a recommendation engine combining **RAG models** with collaborative filtering techniques.
- Leveraged **LLMs** for generating personalized content suggestions.
- Applied prompt engineering to ensure the recommendations were contextually relevant and engaging.
- Pre-processed and curated large-scale datasets to train the **RAG and LLM models**.
- Conducted experiments to **optimize retrieval** and generation components for better user experience.
- Collaborated with data engineers and frontend developers to deploy the recommendation system.
- Developed custom chains using **LangChain** to integrate different AI models and enhance the overall functionality of the AI-powered features.

Technologies Used:

- Programming Languages: Python, JavaScript, Kotlin, sql, pyspark, scala
- Libraries & Frameworks: TensorFlow, Transformers (Hugging Face), React, Node.js, LLM's RAG's, Prompt Engineering
- Tools: Databricks, Git, Docker
- Platforms: AWS for cloud services and deployment

PROJECT4: Internal Chatbot Project: Company Data Retrieval System

Overview: Developed an internal chatbot to streamline access to company data, improving efficiency and productivity by allowing employees to retrieve information and reports through natural language queries.

Key Responsibilities:

- **Project Initiation and Planning:** Collaborated with stakeholders from various departments to identify key requirements and use cases for the chatbot. Developed a detailed project plan, including milestones, timelines, and resource allocation.
- **Data Integration and Management:** Integrated the chatbot with internal databases and data warehouses, ensuring secure and seamless access to company data. Implemented data validation and cleaning processes to maintain data integrity.
- **NLP Model Development:** Built and fine-tuned an NLP model using BERT to understand and process natural language queries. Trained the model on internal documents, reports, and communications to enhance its understanding of company-specific terminology and context. Implemented a named entity recognition (NER) system to accurately identify and retrieve specific entities and data points from the company's databases.
- **Chatbot Development and Deployment:** Designed and developed the chatbot interface using Microsoft Bot Framework and integrated it with Microsoft Teams for easy access by employees. Deployed the chatbot on Azure, ensuring high availability and security through appropriate access controls and monitoring.

- **User Experience and Feedback:** Conducted user acceptance testing (UAT) with employees from various departments to gather feedback and refine the chatbot's functionality and accuracy.
Implemented an iterative feedback loop to continuously improve the chatbot based on user input and evolving requirements.

Technologies Used:

- Programming Languages: Python, JavaScript, Kotlin, SQL, PySpark, Scala
- Libraries & Frameworks: Transformers (Hugging Face), Microsoft Bot Framework, Rasa NLU, LUIS (Language Understanding Intelligent Service) - Microsoft Azure
- Tools: Databricks, Git, Docker, Neo4J, Next.js
- Platforms: Microsoft Azure, Microsoft Teams

Achievements:

- Launched the chatbot, achieving a 95% accuracy rate in understanding and responding to user queries.
- Increased employee productivity by 20% by reducing the time required to access and retrieve information.
- Enhanced data accessibility and transparency, leading to more informed decision-making across the company.
- Received positive feedback from employees, with a 90% satisfaction rate reported in post-launch surveys.
- Integrated **RAG models and LLMs** with existing chatbot frameworks to enhance response generation.
- Employed **prompt engineering** to optimize the interaction flows and improve the quality of chatbot responses.
- Trained and fine-tuned the models using customer interaction logs and FAQ datasets.
- Worked closely with the product team to ensure seamless integration and deployment of the chatbot

Data Visualization & Predictive Analytics

- Steering rapid model creation in **Python** using Pandas, NumPy, SciKit-Learn & plot.ly for data visualization
- Constituting NLP models for **Sentiment Analysis** & MapReduce modules for predictive analytics in Hadoop on GCP
- Designing a real-time contextual behavioral personalization system via econometric & ML to **predict customer behavior**
- Created linear regression model to arrive at a potential transaction price for all future transactions.
- Developed over 50 reports and dashboards on **Power BI** resulting in a 30% increase in operational efficiency.
- Collaborated with cross-functional teams to identify data sources and streamline data collection resulting in a 25% reduction in report development time.
- Led the design and development of Markov model algorithms to enhance predictive analytics within our machine learning platform.
- Collaborated with software developers to optimize the performance and accuracy of Markov models, ensuring seamless integration with existing machine learning workflows.
- Applied Markov models to various real-world scenarios, such as user behavior prediction, system reliability analysis, and financial forecasting, significantly improving predictive capabilities and decision-making processes.
- Worked closely with cross-functional teams to integrate Markov model techniques into end-to-end machine learning pipelines, facilitating smoother data processing and more accurate outcome predictions.
- Conducted in-depth research on advanced Markov processes, including Hidden Markov Models (HMMs) and Markov Decision Processes (MDPs), contributing to the development of innovative solutions and new algorithmic approaches.
- Provided technical leadership and mentoring on Markov models to junior data scientists and developers, fostering a collaborative and knowledge-sharing environment.
- Implemented performance monitoring and reporting mechanisms for Markov model algorithms, utilizing key performance indicators (KPIs) to measure effectiveness and drive continuous improvement.
- Orchestrated & structured data and set up data-access tools effectively for data scientists
- Employing mathematics, statistics, econometrics, and operations research for developing machine learning solutions
- Successfully trained the neural network through unsupervised learning with multiple terabytes of data

Statistical Modeling & ML Algorithms

- Placed various machine learning techniques to build **dynamic pricing** models and maximize profitability
- Designed and created logical and physical data models using **ERwin**, resulting in improved data organization and accessibility for the organization's data-driven initiatives.

- Led the development of a **performance assessment & pricing analysis platform** via **k-NN Algorithm**
- Formed **multivariate regression-based attribution models** using ad stock analysis from the digital marketing data
- Generated **segmentation models** using **K-means Clustering** in order to discover new segments of users

Key Achievements

- **Established the Data Science division from scratch** by recruiting, on-boarding & training a team of 8 Data Analysts.
- Designed and developed reports and dashboards using **Power BI Platform** . Gathered data from various sources including SAP and Excel.
- Proposed a reduction in negotiation time from an average of 3 weeks to 8 days & a 25% jump in closure
- Improved report development through continuous collaboration with both stakeholders and developer teams across different business verticals.
- Consulted over 23 clients on how to develop and maintain their personal self-service dashboard, which resulted in an overall increase of 19% in report accuracy.
- Successful in an overall **loss reduction of 10%** on monthly revenue by implementing the loss minimization techniques
- Migrated data transformation processes on Hadoop to **reduce the data processing time by 25%** .
- Developed a customer **segmentation algorithm** using Python to boost sales leads & increase **market share by 28%**

Data Analytics and Predictive Modelling

- Assisted in developing the Prioritization & Propensity model via Predictive Analysis, logistic regression & other statistical techniques on patients data.
- Processed structured/unstructured data & gained insights by leveraging predictive modelling, statistics, ML & data mining
- Rendering assistance in developing & deploying models while conducting predictive data mining to identify insights .
- Applied Machine Learning algorithms for identifying the most significant features across multiple datasets including medicinal, medical equipments datasets.
- Applied Machine Learning algorithms for identifying the most significant features across multiple datasets
- Utilizing deep learning packages like **Tensorflow, H2O, lighGBM. Theano, Spark MLlib**, etc. to develop ML models
- Applied **Bayesian Model Averaging (BMA)** to combine individual keyword-level models.
- Conducting data analysis to aid in developing models by employing machine learning and artificial intelligence to detect issues

Sentiment Analysis, Clustering, Customer Segmentation, and Generative AI based NLP tasks

- Led a project to develop a conversational AI chatbot powered by OpenAI's GPT-3 language model.
- Developing NLP systems as per the requirements and implementing the right algorithms & tools for NLP tasks
- Deployed statistical and machine learning models using the gathered data to evaluate sentiment and POS tagging via NLP
- Created language models from text data & developed an autonomous Chat-bot based system
- Transforming natural language data into useful features via NLP techniques to feed classification algorithms
- Processed terabytes of data via Machine Learning & Predictive Modelling to develop dynamic solutions.
- Implemented a **convolutional neural network (CNN)** for image classification, achieving an accuracy of 95% on a dataset of 10,000 images.
- Conducting data analysis to aid in developing models by employing machine learning and artificial intelligence to detect issues
- Trained a natural language processing (NLP) model using BERT for sentiment analysis of customer reviews, improving sentiment accuracy by 20%.
- Designed and implemented a text summarization system using OpenAI's transformer-based models.
- Reduced manual content curation time by 40% and improved the accuracy of summaries.
- Leveraged OpenAI's Document Understanding models to automate document classification and data extraction processes.

Mar '18 - Nov '18

BigFoot Retail Solutions Pvt Ltd | Delhi, IN

Logistics Aggregator With a presence in 200+ countries Servicability in delivering Shipments, Shiprocket has delivered cost-effective IT solution

Technology Stack: Python, Pandas, NumPy, Databricks, SciKit-Learn, Lambda, Jupiter Notebook, AWS cloud, SAP HANA ML, Sagemaker, SP HANA

Segmentation & Clustering

- Applied large scale & low latency machine learning for non-parametric models & **high-dimensional data visualization**
- Created **multivariate regression-based attribution models** & segmentation models using **K-means Clustering**
- Utilized high dimensional data sets from users/media agencies/3rd-party apps via PCA, LDA & **Kernel Approximations**
- Designed **ARIMA** to forecast demand and Random Forest model to predict parts at risk of being out of stock
- Assisted in configuring performance trackers, reports, and dashboards for all promotional & marketing activities
- Deployed mathematical, statistical and programming expertise to formulate strategy using a deep learning framework
- Designed a machine learning model via logistic regression to predict consumer complaints
- Creating algorithms based on deep-dive statistical analysis & predictive data modeling to personalize customer interactions

Key Achievements

- Developed an additive scoring model for QSM and a logistic regression model to yield a **K-S statistic of 51.5**
- Deployed **SGD, Logistic Regression, Random Forest, SVM**, etc. for classification models to boost **avg. click rate by 34%** .
- Assisted the senior business analysts in devising statistical models for forecasting quarterly sales figures.

Nov '16 - Feb '18

Data Analyst

Netway India Pvt Ltd | Gurugram, IN

India's First Doctor's App and Astrology Service App having huge customer base across Globe.

Data Analytics & Model Development

- Directed model development, validation, testing and implementation of analytical products and applications
- Suggested tweaks and new features for improving engagement by 29% & user retention by 34%.
- Forecasted the quarterly sales figures and revenues to devise new strategies for the company using **ARIMA, ARMA, SARIMAX** and other Time-Series Models.
- Configured performance trackers, reports, and dashboards for all promotional activities.

ML Algorithms & Statistical Analysis

- Stationed advanced text mining algorithms to identify search intent latent in individual keywords.
- Tested and implemented decision trees, random forests and ensemble models using bagging and boosting
- Utilized hypothesis-driven approach to analyze A/B testing results & offer recommendations based on the test performance
- Employed Principle Component Analysis to analyze collinearity and reduce the dimensionality of the dataset

Apr '15 - Oct '16

Data Analyst

Corwhite Solutions Pvt Ltd | Gurugram, IN

Cloud Taxation and E-Filings of Income Tax and other Consulting Services

Data Analytics & Model Development

- Directed model development, validation, testing and implementation of analytical products and applications
- Designed database cubes successfully and star schemes using ER/Studio .
- Monthly sales from new customer in the US and other countries using Time-Series Modelling using different models like **ARMA, ARIMA, SARIMAX** etc

- Monitoring key metrics & understanding root causes of changes in metrics to identify opportunities for improvement.
- Generating a monthly report on user behavior for the seniors within a stipulated time frame while ensuring accuracy.

ML Algorithms & Statistical Analysis

- Stationed advanced text mining algorithms to identify search intent latent in individual keywords.
- Tested and implemented decision trees, random forests and ensemble models using bagging and boosting.
- Employed Principle Component Analysis to analyze collinearity and reduce the dimensionality of the dataset.

EDUCATION

B.TECH - Electronics and Communication Engineering

Aug '11 - Apr '15

Dr.K.N.M.I.E.T | Delhi, IN

Dr.KNMIET is the very first and renowned collage affilated to Dr.A.P.J Abdul Kalam Technical University

Master's - Artificial Intelligence and Machine Learning

Jan '23 - Feb '24

BITS PILANI | Pilani, IN

Bits Pilani is one of the Top Ranked University in India

ADDITIONAL INFORMATION

- Languages: English, Hindi

AREAS OF EXPERTISE

